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Improving Raman spectroscopic differentiation of the geographical origin of rice by simultaneous illumination over a wide sample area

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This research substantially improved the differentiation of rice of two geographical origins utilizing a wide area illumination (WAI) scheme capable of collecting Raman spectra of a large sample area (28.3 mm^2) synchronously without sample rotation. For the purposes of comparison, we also employed a conventional scheme in which the laser illuminated only small areas. Principal component analysis (PCA) was used to differentiate the two geographical origins using the Raman spectra collected through both the conventional and the WAI schemes. The WAI scheme exhibited improved differentiation, primarily due to the fact that the WAI scheme could efficiently produce Raman spectra with a more reliable sample representation, as well as better reproducibility. The spectral feature obtained using the conventional scheme appeared to be more variable; however, this variation resulted more from unsuccessful sample representation than from an actual change in the sample composition. We conclude that the WAI scheme has a good potential for the analysis of diverse agricultural samples that consist of solid granules. **Copyright © 2008 John Wiley & Sons, Ltd.**

Keywords: wide area illumination Raman scheme; geographical origin discrimination; moving window PCA; rice

Introduction

Previous research to determine the geographical origins of diverse agricultural products has focused especially on the utilization of diffuse reflectance near-infrared (NIR) spectroscopy.[1–10] As most of these products are solids in the granular form, it is easy to collect the diffuse reflectance NIR spectra with minimal sample preparation while a sample is rotating or reciprocating to cover a large surface area for accurate sample representation. In addition, a granular sample can be ground into a fine powder prior to spectral collection so as to further improve both sample representation and spectral reproducibility. Although the NIR spectral features are not highly distinct because of the nature of overtone and combination bands, its speed and simplicity make it the most frequently selected spectroscopic method for the determination of the origins of agricultural products.

Raman spectroscopy is also capable of collecting data quickly and providing more selective spectral features than NIR spectroscopy; however, it has not been as commonly utilized. Only a few researches describing the use of Raman spectroscopy to determine the origins of agricultural products have been published.[10–13] One of the reasons for this is the fact that the laser illuminates only a small spot (50–500 μm) when collecting spectral information, thereby increasing the chances of inaccurate compositional sample representations. In addition, Raman intensity could be sensitive to the variations of surface morphology and sample orientation. These problems can be overcome by sample rotation during Raman collection, so as to cover a wider sample area and to represent the average sample features, rather than just a localized feature.[14–17] To make Raman spectroscopy more competitive for the analysis of agricultural products, where most of the samples are of *large grain* forms, it is necessary to devise an efficient spectral collection scheme capable of representing correct sample identities without having to rotate the samples. Sample rotation during spectral collection leads to longer analysis times and requires a cumbersome external mechanical device. In addition, when a continuous online measurement is targeted, a Raman collection scheme with sample rotation cannot be adopted. We have recently presented the applications of a wide area illumination (WAI) Raman scheme for the analysis of various pharmaceutical products.[18–21] Using a laser capable of illuminating a large sample area (28.3 mm^2) for Raman spectral collection (a 6-mm laser spot with a focal length of 248 mm), the sample representation and reproducibility of Raman measurements improved considerably. The same WAI Raman scheme improved the robustness of Raman measurement compared to the conventional small-spot schemes for inline monitoring of the wet granulation process.[22] Addi-

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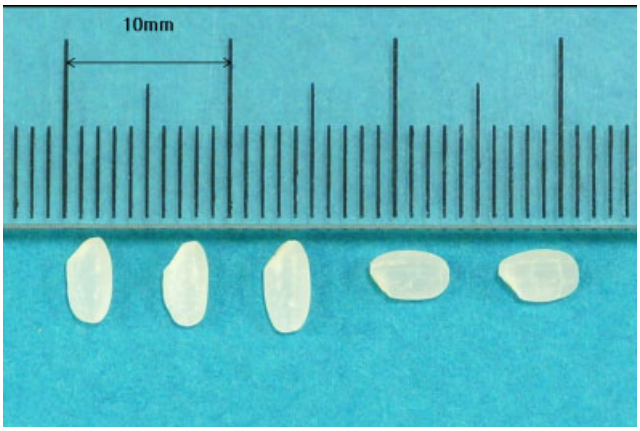

Figure 1. Picture of five different rice grains with a relevant scale. This figure is available in colour online at www.interscience.wiley.com/journal/jrs.

tionally, it has demonstrated the capability to perform mapping through several layers of polymers up to 13 mm.[23]

Alternatively, to improve the sample representation, spatially offset Raman spectroscopy (SORS) and transmission Raman spectroscopy (TRS) have been extensively studied and the corresponding applications for living tissue and powders have been reviewed.[24] These methodologies were able to successfully produce a Raman spectrum representing chemical identity in the inner (deeper) part of a solid sample and reduce the superimposed fluorescence background simultaneously. The samples studied for SORS and TRS were mostly fine powders rather than large size granules with irregular shape such as whole grain samples.

In this research, we evaluated a WAI Raman scheme with the goal of identifying rice of two different geographical origins. The rice samples used were whole grains that were not further ground. The grains were predominantly oval shaped (Fig. 1). The average lengths of the major and minor axis, based on measurements from 20 independent rice grains, were 4.89 ± 0.23 mm and 2.82 ± 0.07 mm with thicknesses of 2.04 ± 0.08 mm, respectively. As the size of the rice grains is relatively large, it requires a Raman collection scheme that covers a wide area. If a laser illuminates only a tiny spot on a rice grain, only localized spectral information can be acquired.

For comparison, both the WAI and conventional Raman collection schemes were used to collect spectra of all grains. The size of the laser spot for the conventional collection scheme was approximately 80 μm. Principal component analysis (PCA)[25] was used to convert the resulting Raman spectral features into fewer dimensions of scores. The Raman spectral features of the two rice types originating from different locations were very similar. Only a minor spectral difference could be recognized for differentiation; therefore, correct sample representation and improved spectral reproducibility are critical factors to determining the success of identifying the geographical origins of rice.

Experimental

All the 30 samples of Chinese and Korean rice were obtained from the National Agricultural Products Quality Management Service (NAQS) in Seoul, Korea. The samples composed of different species and were cultivated in diverse geographical regions. These samples were prepared over the course of 1 year to include the wider compositional variations in a dataset. The grain samples shown in Fig. 1 were used immediately for the collection of Raman spectra, without further sample preparation.

The schematic diagram of the WAI scheme is explained in detail in a previous publication.[18] The WAI scheme used in this study has been developed by Kaiser Optical Systems (Ann Arbor, MI, USA). An excitation laser (785 nm, diode laser) was magnified to form a circular illumination area with a diameter of 6 mm (area 28.3 mm^2), so as to cover a large sample area. The scattered radiation was then collected by an array of 50 optical fibers and delivered to the charge-coupled device (CCD) detector (Kaiser RamanRxn1 unit; Kaiser Optical Systems, Ann Arbor, MI, USA). The radiation from the optical fibers was dispersed via a holographic transmission grating. Then a combined signal was integrated over the total illuminated area. The rice grains were transferred to a 1-cm quartz cuvette located at the focal point of the laser. Simultaneously, Raman spectra of the rice samples were collected using a noncontact optical fiber probe (Kaiser Optical Systems), yielding a laser spot of approximately 80 μm with a focal length of 64 mm. For both cases, each Raman spectrum corresponded to an accumulation of 64 scans, with an exposure time of 2 s for each scan.

Triplicate spectra were collected for each sample. At each spectral collection, the quartz cuvette containing the rice samples was shaken vigorously to redistribute the grains, thereby ensuring that the sample orientation at each measurement would be random. PCA and moving window principal component analysis (MWPCA)[23–25] were performed using Matlab Version 7.0 (MathWorks Inc., MA, USA).

Results and Discussion

Examination of spectral features

Prior to examining spectral features or performing further spectral processing, all the spectra were baseline-corrected and normalized. Figure 2(a) shows the 30 raw Raman spectra from 10 of the samples (randomly selected) collected using the conventional scheme. The most noticeable feature is the baseline variation among spectra. As the surface of the rice grain is round and irregular, the location of a grain with respect to the focal point will not be consistent in the measurements. This leads to considerable baseline variation. As there was no nonlinear background superimposed in the sample spectra such as fluorescence, the baselines of spectra were linearly offset to zero at four different wavenumbers (1500, 813, 692, and 400 cm^{-1}) and the peak area under the 1500–400 cm^{-1} range was calculated. Then, each baseline-corrected spectrum was divided by the corresponding peak area under the 1500–400 cm^{-1} range. Through normalization, the possible Raman peak intensity variation resulting from a change in sample position with respect to the focal point could be greatly reduced. The resulting normalized spectra are shown in Fig. 2(b). For further spectral comparison and PCA, normalized spectra were used throughout this study. For Raman spectra collected using the WAI scheme, the same normalization procedure was undertaken for further spectral processing.

Figure 3(a) shows the Raman spectra of two randomly selected samples (one from Korea, one from China) collected by using the conventional scheme. The overall spectral features of both the samples are almost identical; however, there are some minor spectral differences in the 1450–900 and 650–450 cm^{-1} range. Rice is composed of approximately 80–84% starch and

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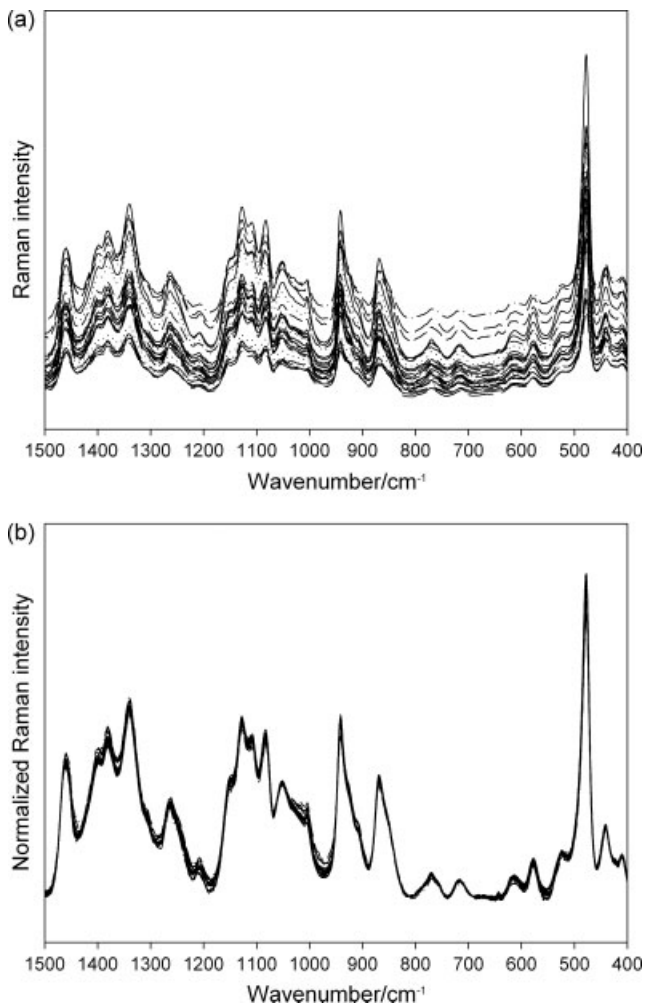

Figure 2. Thirty raw Raman spectra from 10 samples (randomly selected) collected through the conventional scheme (a) and the corresponding normalized spectra (b).

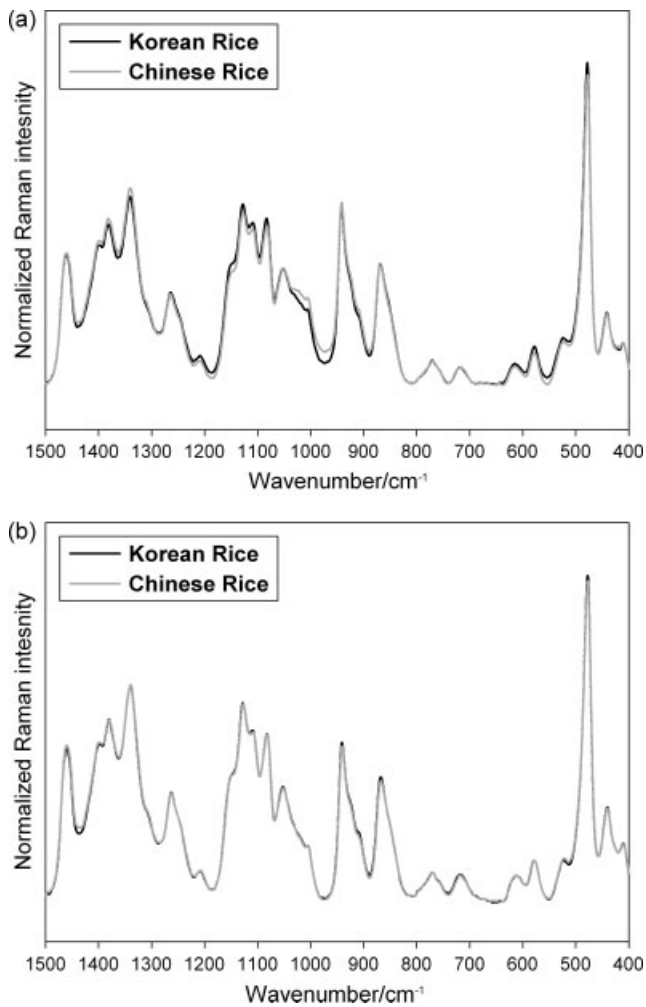

Figure 3. Two Raman spectra of two samples (one from Korea, one from China) collected using the conventional (a) and WAI schemes (b).

7–9% protein. Fats, minerals, and vitamins are the other minor constituents. Therefore, the resulting Raman features are obviously dominated by those of starch. As starch is a huge polymer of a glucose unit, with a molecular weight ranging from 50 000 to 200 000 Da, it would be hard to find spectral differences in starches. Conversely, although the protein content is much lower, this could also provide valuable spectral clues for differentiating between rice samples of different geographical origins, because of the higher variability in the chemical structure. Figure 3(b) details two Raman spectra of two randomly selected samples collected using the WAI scheme. As before, the overall spectral features were similar to each other; however, tiny spectral differences were observed in the 1500–1350, 1000–850, and 750–500 cm^{-1} range. Overall, it appeared that the Raman spectra collected using the conventional scheme provided more discernable spectral differences than those collected using the WAI scheme; however, these spectral variations should be based on changes in structure and composition rather than on unsuccessful sample representations or degraded spectral reproducibility.

The spectral difference between Chinese and Korean rice is very small (Fig. 3); therefore, the critical factor that can differentiate the Chinese *versus* Korean rice is correct sample representation. If the magnitude of the spectral variation from an unsuccessful sample representation is larger than that of the spectral variation from real compositional and structural difference, it will be difficult to differentiate between Chinese and Korean rice.

Principal component analysis

Initially, using all spectra (90 spectra from each origin, 180 spectra total) PCA was performed over the full spectral range (1500–400 cm^{-1}). We first generated six scores, later selecting the best combination of two scores through an evaluation of identification errors at the given score combinations. A combination of two scores was used because no further significant improvement in differentiation was achieved by adding more scores (dimensions). Additionally, it was easy to visualize the performance of differentiation in the domain of two-dimensional score distribution.

We calculated the identification error using the following equation, when two groups were partially separated:

$$\textit{Identification error} = \frac{\sum_{n=1}^{N} XI_n}{\sum_{n=1}^{N} XI_n + \sum_{n=1}^{N} XC_n} \times 100\% \qquad (1)$$

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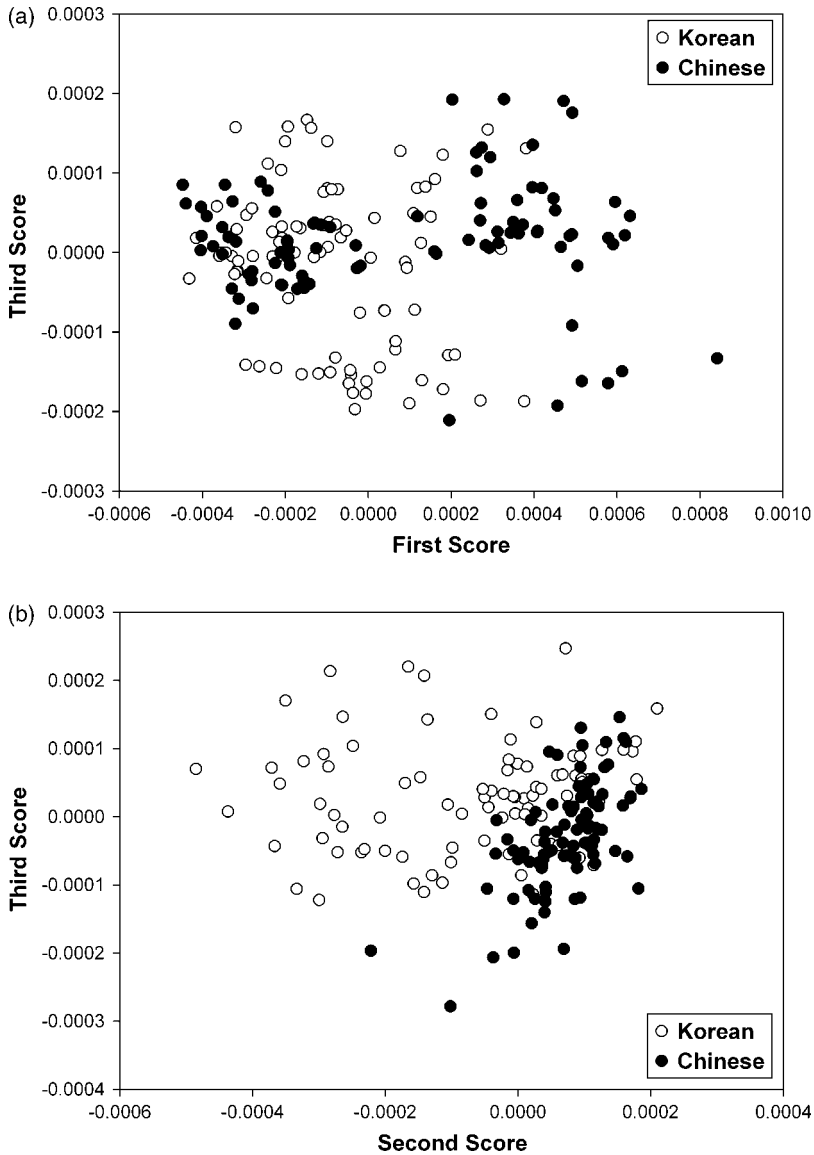

Figure 4. Two-dimensional score scatter plots generated using the conventional (a) and WAI schemes (b). The full spectral range (1500–400 cm^{-1}) was used for PCA.

where n is the number of spectra. Initially, the center points of each group were calculated in a two-dimensional score space and then a line perpendicular to the line between the two center points was defined as the borderline between the two groups. The borderline was located at the middle of the line between the two center points. After the borderline was established, the perpendicular distances from each sample point to the borderline were calculated (XI_n, distance for incorrectly assigned sample n; XC_n, distance for correctly assigned sample n). Finally, the identification error was obtained by calculating the percentage of the sum of incorrectly assigned samples over the total sum of distances (correctly and incorrectly assigned samples), as shown in Eqn (1).

The resulting score scatter plots from the conventional and WAI schemes are shown in Fig. 4. The full spectral range (1500–400 cm^{-1}) was used in both cases. Filled and open circles correspond to Chinese rice and Korean rice, respectively. Additionally, the identification errors and scores used are summarized in Table 1. When the conventional scheme was used, the scores of Chinese and Korean rice were widely spread, making it difficult to identify members belonging to each of the two groups. Therefore, the spectral differences observed in Fig. 3(a) might not fully originate from a difference in chemical identities and the differences might originate instead from an incorrect sample representation by the conventional scheme. Conversely, two groups of Chinese and Korean rice were identifiable, although there was some overlap between the two groups. Differentiation between geographical origins is greatly improved by using the WAI

Table 1. The results of PCA using the full spectral range. The whole spectral dataset was used directly for PCA

Measurement scheme	Spectral range (cm^{-1})	Scores used	Identification error (%)
Conventional	1500–400	1, 3	32.8
WAI	1500–400	2, 3	12.5

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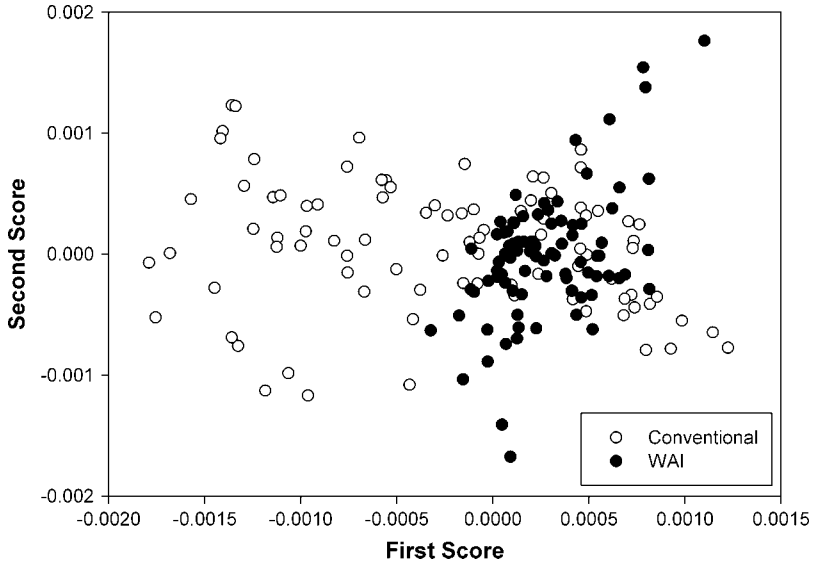

Figure 5. Score scatter plot generated using difference spectra from the same samples. Open and filled circles correspond to the conventional and WAI schemes, respectively.

Raman scheme. This could result from the fact that WAI is superior in representing correct sample compositions without localized features and spectral reproducibility, as previously argued.[18]

To investigate the difference in sample representation between the conventional and WAI schemes, we examined triplicate spectra (normalized spectra) collected for the same sample. If spectral variation existed among triplicate spectra, this would be indicative of an unsuccessful sample presentation due mainly to sample inhomogeneity relative to the small size of the laser spot and unsatisfactory spectral reproducibility. On the basis of a simple visual observation of triplicate spectra, it was difficult to find visually noticeable differences. To characterize unforeseen differences among triplicate spectra, the difference spectra were generated by subtracting one spectrum from another. Thus, three difference spectra were able to be generated from the triplicate spectra for one rice sample, and 90 difference spectra were obtained from each geographical origin. As the difference spectra were obtained from the triplicate spectra of the same sample, the difference should ideally be close to zero.

PCA was performed using the difference spectra from the conventional and WAI schemes using the whole spectral range of 1500–400 cm^{-1}. Figure 5 details the resulting score scatter plot from the first and second scores. Many scores are centered near zero; however, the distribution of scores from the conventional scheme is wider than that from the WAI scheme, especially in the domain of the first score. This is a clear indication that the difference among triplicate spectra of the same sample is larger when the conventional scheme is used. This resulted from the inconsistent representation of samples due to the fact that the laser illuminates only a tiny spot when collecting spectral information. When a wider area is illuminated, as in the WAI scheme, the score distribution of difference spectra is more aggregated. This indicates that the sample representation by the WAI scheme is more consistent and reproducible. Consequently, better separation between the two rice groups using the WAI scheme (Fig. 4) is largely attributed to the correct and reproducible sample representation. This fact becomes much more important when minute spectral differences must be recognized, as was the case for this study.

Table 2. The found optimal spectral ranges and corresponding results by the cross-validated MWPCA

Measurement scheme	Spectral range (cm^{-1})	Scores used	Identification error (%)
Conventional	1500–1350	1, 2	14.0
WAI	1475–1375	1, 2	4.1

Moving window principal component analysis

The use of the whole spectral range for PCA may not be desirable if the spectral difference between Chinese and Korean rice is small and localized. Because an optimal spectral range may exist that yields improved differentiation results, in this study we used MWPCA in an attempt to find the optimal spectral range.[26–28] The main strategy of MWPCA is to perform PCA at many different spectral ranges by changing the size and location of the spectral window, thereby determining the spectral range that yields a minimal error. The window size was varied from 50 to 1100 cm^{-1} at increments of 50 cm^{-1}. For every given window size, the window was shifted from low to high wavenumbers in 25 cm^{-1} steps until the whole spectral range (1500–400 cm^{-1}) was covered. This method enabled a total of 253 ranges to be evaluated.

To improve the statistical significance of the resulting identification errors, we cross-validated the whole dataset by dividing it into five segments (12 samples in each segment, 6 samples from each origin). For every given range, a PCA model was built using four segments, with one remaining segment being predicted, and the corresponding identification error rate was calculated. This procedure was repeated five times until all the samples were predicted. Finally, the average identification error was obtained. Table 2 summarizes the results of MWPCA.

In both cases, the identification errors were improved when an optimized spectral range was used. The optimized spectral ranges for the use of conventional and WAI schemes are 1500–1350 cm^{-1} and 1475–1375 cm^{-1}, respectively, with the resulting identification errors being 14.0% and 4.1%, respectively. Although the identification errors differ significantly from each

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other, the resulting optimal ranges are generally similar. It would be difficult to extract a reasonable relationship between the found optimal range and the corresponding spectral features because of the minor spectral differences between the rice samples from the two regions; however, the identification error was significantly improved using the WAI scheme. This implies that the Raman spectra obtained using the conventional scheme are less able to represent correct sample identity and less reproducible for solid granular samples such as rice.

Conclusions

Wide sample area coverage to secure correct sample representation and reliable analytical result is a well-known necessity in spectroscopic analysis of solid samples such as rice grain in this study. To meet the requirement of correct sample representation, the use of WAI scheme would be more advantageous and simpler over the conventional approach, requiring an external device for sample rotation during Raman collection. Raman spectra collected using the WAI scheme were representative of the rice samples, although these were large whole grains without any further grinding or sample rotation. Although the number of samples used in the study was not large, we have shown the practical potential of Raman spectroscopy for the geographical differentiation of agricultural products including rice, where NIR spectroscopy has been predominantly utilized. Future research will seek to perform compositional analyses of agricultural products using the WAI Raman scheme, such as the determination of protein and fat content, where correct sample representation and reproducibility are more critical.

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